

VICKY JEBET

GIS and Remote Sensing Analyst | MEL Practitioner | Impact Data Specialist

Nairobi, Kenya | jebetv15@gmail.com | <https://jebetvicky.github.io>

PROFESSIONAL SUMMARY

Geospatial Data Specialist and MEL Practitioner with 3+ years of professional experience translating Earth Observation data and spatial analysis into actionable intelligence for food security, climate resilience, environmental conservation, and sustainable development across East and Horn of Africa. Combines technical expertise in GIS, remote sensing, and geospatial programming (Python, R, Google Earth Engine) with strong capacity in Monitoring, Evaluation and Learning (MEL), program design, and stakeholder engagement. Currently completing MSc in Agricultural Information and Communication Management while advancing MEL competencies through the AWARD Young Impact Associate Fellowship focused on agrifood systems and gender-responsive evaluation. Demonstrated impact across humanitarian, development, climate finance, and research contexts, with publications in peer-reviewed journals and proven ability to build technical capacity among diverse stakeholder groups. Passionate about leveraging space technology and data science to support climate action, food security, and equitable development in Africa.

CORE COMPETENCIES

Geospatial Technologies: QGIS (Advanced) | ArcGIS Suite | Google Earth Engine (Python) | Digital Earth Africa | SEPAL | Sentinel Application Platform (SNAP) | ERDAS Imagine

Remote Sensing & Analysis: Satellite Imagery (Landsat, Sentinel-2, Sentinel-1, MODIS) | Spectral Indices (NDVI, NDWI, VCI) | Change Detection | Damage Assessment | LULC Classification | Deforestation Tracking | Flood Risk Analysis | Vegetation Monitoring

Data Science & Programming: Python (gdal, rasterio, geopandas, earthengine-api, pandas, numpy) | R (tidyverse, ggplot2, raster, sf) | Bash Scripting | Data Pipeline Development | Spatial SQL (PostGIS/PostgreSQL) | Automation & Workflow Optimization

Data Visualization & Analytics: Tableau (Advanced) | PowerBI | Matplotlib | Seaborn | Web Mapping Services | Interactive Dashboards | Cartography & Infographics | Spatial Data Exploration

MEL & Impact: Monitoring, Evaluation & Learning Frameworks | Results-Based Management | Impact Measurement & Learning | GESI Integration | Indicator Development | Participatory Evaluation | Decolonising Evaluation Approaches | Data Quality Assurance

Field & Stakeholder Engagement: GPS Data Collection (KoboToolbox, ODK, Survey123) | Field Validation & Ground-Truthing | Participatory Mapping | Capacity Building & Training Facilitation | Stakeholder Coordination (11+ countries) | Community Sensitization | Multi-disciplinary Team Collaboration

Project & Program Support: Project Management | Technical Documentation | Data Acquisition & Preprocessing | Metadata Management | Quality Control Processes | Workflow Documentation | Remote Collaboration & Distributed Teams | Knowledge Translation & Science Communication

PROFESSIONAL EXPERIENCE

MEAL Young Impact Associate

ICIPE (International Centre of Insect Physiology and Ecology) — BioInnovate Africa Programme | June 2026 – November 2026 | Nairobi, Kenya |

Under mentorship of Dr. Michael Kidoido, supporting BioInnovate Phase II partner follow-up survey, fellow tracer study, and MEL integration for BioInnovate III

- Supporting BioInnovate Africa Programme with geospatial data collection, analysis, and MEL documentation
- Conducting follow-up survey with BioInnovate Phase II partner organizations to assess project outcomes and sustainability

- Contributing to tracer study tracking career trajectories and impact of BioInnovate fellows
- Developing M&E support and data-driven learning systems for BioInnovate III program design
- Integrating GESI and participatory evaluation approaches into program monitoring frameworks
- Leading G-MiP (GESI MEL Mini-Project) scoping and implementation

GIS Analyst

Earth Banc: Earthtree Kenya (Regenerative Kenya Project) | May 2025 – October 2025 | Trans Nzoia County, Kenya |

- Led geospatial analytical workflows for Gold Standard-certified ARR/FMNR carbon project covering 600+ hectares
- Conducted LULC classification, slope/elevation modelling, historical deforestation tracking, and restoration suitability mapping using Sentinel-2 and Landsat time-series data with Python automation
- Produced detailed analytical maps integrating terrain, soil, and biomass datasets to inform project baseline definition, additionality assessments, and land eligibility verification for third-party carbon credit validation
- Digitized and structured 64+ cadastral and registry maps into unified database, reducing data retrieval time by 40% and streamlining land eligibility verification workflows
- Designed spatially optimized restoration layouts for 600+ hectares using raster analysis and spatial optimization, reducing planting errors by 29% and improving carbon sequestration projections
- Conducted field validation across 422+ hectares using GPS-based ground-truthing and change detection verification, achieving 93%+ spatial accuracy and ensuring Gold Standard third-party verification readiness
- Developed and documented GIS Standard Operating Procedures aligned with Gold Standard carbon methodology requirements, fostering workflow consistency, reproducibility, and long-term audit readiness
- Collaborated remotely with field teams, technical experts, and community stakeholders to validate analytical outputs, communicate findings, and ensure alignment with operational realities

GIS Analyst

IGAD Climate Prediction & Applications Centre (ICPAC) | February 2023 – December 2024 | Nairobi, Kenya

- Coordinated geospatial support for food security, climate resilience, and early warning systems across 11 IGAD member states (Kenya, Ethiopia, Tanzania, Uganda, Somalia, South Sudan, Sudan, Eritrea, Rwanda, Burundi, Djibouti)
- Processed and analyzed multi-temporal satellite imagery (Landsat, Sentinel-2, MODIS) using Google Earth Engine and geospatial Python workflows to generate actionable LULC insights and vegetation stress assessments
- Applied spectral indices (NDVI, NDWI, VCI) and climate indicators (precipitation anomalies, land surface temperature trends) to monitor vegetation conditions, agricultural productivity, and ecosystem dynamics across Horn of Africa
- Developed, maintained, and optimized GMES Data Center, ensuring timely and accurate provision of Earth Observation datasets to regional stakeholders and improving request fulfillment times by 28%
- Designed and maintained IGAD Resilience Measurement Dashboard in Tableau, delivering real-time visualization of 15+ resilience indicators supporting climate risk planning, early warning systems, and disaster risk reduction decision-making
- Produced thematic maps, infographics, interactive web maps, and spatial datasets for projects addressing biodiversity conservation, food security, agriculture, natural resource management, and environmental monitoring
- Coordinated GEOGLAM Crop Monitor Learning Exchange, fostering international collaboration on satellite-based agricultural monitoring and food security mapping across 11 countries
- Facilitated 3 regional capacity-building workshops delivering hands-on GIS and remote sensing training to 60+ professionals from national agencies, government departments, regional organizations, and partner institutions

- Participated in Greater Horn of Africa Climate Outlook Forum as rapporteur, documenting technical discussions, synthesizing seasonal climate forecast insights, and supporting dissemination to regional stakeholders
- Co-authored 'Forecasting and Monitoring Extreme Events Over Africa' (Elsevier, 2025), contributing geospatial analysis, Earth Observation methodology, and climate insights to peer-reviewed research advancing evidence-based environmental monitoring
- Supported GMES & Africa project across biodiversity conservation, food security, agriculture, and natural resource management portfolios using remote sensing-based monitoring and spatial analysis

School Lead Educator

TechLit Africa | September 2022 – February 2023 | Nairobi, Kenya |

- Mentored and taught digital literacy skills to primary school students, fostering confidence, engagement, and digital citizenship
- Cultivated cultural exchange through storytelling and facilitated sense of global community among students
- Maintained computer lab functionality to ensure uninterrupted learning experiences and technical reliability
- Collaborated with grassroots organizations to develop and implement digital literacy programs aligned with community needs and educational priorities
- Supported capacity-building initiatives by training local educators in effective technology adoption and digital pedagogies
- Facilitated events engaging students, parents, and community stakeholders in adopting digital tools for education and economic empowerment
- Coordinated content development and impact reporting on program outcomes, fostering stakeholder trust and program sustainability

GIS & Remote Sensing Intern

Kenya Space Agency | January 2022 – June 2022 | Nairobi, Kenya |

- Conducted advanced LULC mapping and flood risk analysis using Google Earth Engine, producing hazard maps adopted by local disaster management teams in 3+ counties for disaster preparedness and response planning
- Processed multi-temporal Landsat and Sentinel imagery applying spectral indices for vegetation health monitoring, forest cover change detection, water extent mapping, and environmental assessment
- Collaborated with Digital Earth Africa to produce national-scale forest and wetland datasets contributing to regional environmental monitoring and biodiversity conservation initiatives
- Designed and delivered STEM and GIS awareness training to 200+ high school and university students through Kenya Space Clubs, promoting careers in geospatial sciences and space technology
- Coordinated educational outreach programs connecting youth with hands-on learning experiences and career guidance in space technology and geospatial sciences
- Participated in Kenya Space Expo 2022 as planning team member and presenter, fostering public engagement and awareness of space technology applications for development
- Delivered technical presentations showcasing Earth Observation-based solutions for environmental monitoring and disaster risk reduction, resulting in 2 new inter-agency collaborations

EDUCATION & PROFESSIONAL DEVELOPMENT

MSc in Agricultural Information and Communication Management (In Progress)

University of Nairobi | October 2025 – 2027 | Nairobi, Kenya |

Focus: Integrating geospatial technologies, MEL frameworks, and digital communication approaches to strengthen farming systems resilience, support rural livelihood diversification, and improve agricultural knowledge dissemination in African contexts.

Young Impact Associate Fellowship (Cohort 4)

African Women in Agricultural Research and Development (AWARD) | December 2025 – December 2026 | Nairobi, Hybrid | u

Building expertise in Monitoring, Evaluation and Learning (MEL) for agrifood systems and climate resilience initiatives across Africa

- Strengthening MEL competencies for food security and climate resilience programs through evidence-based learning
- Integrating Gender Equality and Social Inclusion (GESI) into MEL frameworks, results-based management, and impact measurement for agrifood systems
- Exploring decolonising evaluation approaches emphasizing participatory methods, locally grounded knowledge, and context-sensitive evaluation in agrifood development

Bachelor of Technology – Geoinformation Technology

The Technical University of Kenya | 2017 – 2022 | Nairobi, Kenya |

Thesis: Monitoring Land Use and Land Cover Change Using Remote Sensing and GIS in South Nandi Forest, Nandi County

Coursework: Geospatial Information Systems | Remote Sensing Techniques | Geospatial Data Acquisition | Digital Cartography | Programming for GIS Applications | Spatial Analysis | Geographic Database Management

Diploma in Information Technology

St. Paul's University | 2018 – 2021 | Nairobi, Kenya |

Remote Sensing Training Workshop Certificate

USAID / USGS / Government of Kenya | 2023 |

Outstanding participation in Level 2 & 3 remote sensing techniques and geospatial analysis

PUBLICATIONS, RESEARCH & THOUGHT LEADERSHIP

Peer-Reviewed Publications:

Gudoshava, M., Chinyoka, S., Mwanthi, A., Koech, E., **Jebet, V.**, Omeny, M., & Otieno, R. (2025). Chapter 12 — [Forecasting and monitoring of extreme rainfall over Africa. *Rainfall Extremes: Understanding, Forecasting, and Adaptation*.](#)

Articles & Knowledge Products:

(2024) ['The Horn of Africa's Blue Carbon Heroes: Seaweeds and Mangroves in the Fight Against Climate Change'](#) — Focus on coastal ecosystem conservation, carbon sequestration, and climate mitigation

(2023) ['Pursuing Sustainable Development Goal 15: Progress and Challenges in Protecting Life on Land'](#) — SDG 15 implementation analysis for terrestrial ecosystem protection

(2023) ['Youth Solutions for Food and Water Security'](#) — Twitter Space Event Co-host and contributor, exploring youth-led innovation in food and water security

VOLUNTEERING & COMMUNITY LEADERSHIP

Member & Youth Mentor

Koibatek University Students Association (KOUSA), Baringo County | September 2017 – Present | Koibatek District, Baringo County, Kenya |

- Designed and led youth mentorship programs for primary and high school students in Koibatek District, building leadership potential, educational outcomes, and economic awareness
- Organized agricultural and environmental conservation campaigns contributing to planting of 3,000+ trees across Koibatek region, engaging youth in sustainable land management and enterprise practices
- Mobilized community resources and coordinated fundraising initiatives for vulnerable community members, demonstrating grassroots leadership and social accountability

- Developed educational materials for youth focusing on climate change, sustainability, environmental conservation, and community enterprise development

Program Facilitator & Mentor

Kuwa Mwenyewe Mentorship Program, East Pokot, Baringo | December 2016 – Present | East Pokot, Baringo County, Kenya |

- Facilitated leadership and character development mentorship sessions for children and youth in East Pokot using Christian values-based approaches and participatory methods
- Organized talent development and economic empowerment workshops for women and youth, promoting entrepreneurship, financial literacy, and livelihood diversification
- Conducted motivational speaking sessions to increase school enrollment and foster education culture in underserved Baringo communities
- Trained and guided youth mentors on mentorship models, program facilitation, and community engagement, ensuring sustainable program delivery and local ownership
- Managed program coordination, scheduling, resource allocation, curriculum development, and documentation for ongoing mentorship cycles

TECHNICAL SKILLS - DETAILED

Geospatial Software & Platforms:

QGIS (Advanced — vector/raster analysis, cartography, scripting, plugins) | ArcGIS Pro & ArcMap | Google Earth Engine | Digital Earth Africa (Sandbox) | SEPAL (Copernicus platform) | Sentinel Application Platform (SNAP) | ERDAS Imagine | QGIS Python Console

Remote Sensing & Imagery Analysis:

Landsat 8/9 (multispectral analysis, time-series) | Sentinel-2 (10m resolution, vegetation indices) | Sentinel-1 (SAR, radar imagery) | MODIS (coarse resolution, vegetation monitoring) | Spectral Index Computation (NDVI, NDWI, NDBI, NDMI, VCI) | Change Detection (differencing, classification comparison) | LULC Classification (supervised/unsupervised) | Atmospheric Correction | Mosaicking & Preprocessing

Programming & Scripting:

Python: gdal, rasterio, geopandas, shapely, pandas, numpy, matplotlib, seaborn, earthengine-api, scikit-learn | R: tidyverse, ggplot2, raster, sf, leaflet, shiny | Bash scripting for automation | Data pipeline development & workflow optimization | Jupyter Notebooks & RMarkdown for reproducible analysis

Data Management & Databases:

PostgreSQL with PostGIS for spatial databases | SQLite for lightweight spatial data | Spatial joins, buffering, topology operations | Metadata management (ISO 19139) | Data versioning & documentation | ETL workflows

Data Visualization & Dashboards:

Tableau (Advanced — interactive dashboards, geospatial visualization, real-time data integration) | PowerBI | Web Mapping Services (WMS, WFS, GeoJSON) | Leaflet.js for interactive web maps | Matplotlib & Seaborn for scientific visualization | Infographics & cartographic design

Field Data Collection:

KoboToolbox (survey design, data aggregation) | ArcGIS Survey123 | ODK (Open Data Kit) | GPS data collection & validation | Ground-truthing protocols | Spatial accuracy assessment

Analysis & Modeling:

Spatial analysis (buffer, overlay, intersection, union) | Raster math & map algebra | Zonal statistics | Accuracy assessment & error matrices | Damage assessment workflows | Suitability mapping | Cost-distance analysis

Collaboration & Productivity:

Slack, Microsoft Teams, Zoom, Google Workspace | Remote collaboration platforms | Version control (Git/GitHub basics) | Project management tools (Monday.com, Trello) | Microsoft Office Suite (Word, Excel, PowerPoint)

LANGUAGES

Kiswahili (Native) | English (Fluent — Native Proficiency) | Kalenjin (Native) | French (Intermediate — A2/B1)

SECTORS & FOCUS AREAS

Primary Sectors: Food Security | Climate Resilience & Adaptation | Sustainable Agriculture & Agrifood Systems | Environmental Conservation & Biodiversity | Land & Water Management | Disaster Risk Reduction | Livelihoods & Community Development

Geographic Focus: East Africa (Kenya, Uganda, Tanzania, Rwanda) | Horn of Africa / IGAD Region | Sub-Saharan Africa

Technical Specializations: Satellite-Based Monitoring | Agricultural Monitoring & Food Security | Climate Data Analysis | Carbon MRV (ARR, FMNR, reforestation) | Disaster Risk Reduction | Environmental Change Detection | Water Resources Management | Wetlands & Blue Carbon Conservation

REFERENCES

Available upon request